3.-

$$C = \frac{2\pi\epsilon_0 l}{\ln\left(\frac{b}{a}\right)} \qquad b = 4a$$

$$C_0 = \frac{2\pi\epsilon_0 l}{\ln(4)} \quad \xrightarrow[\substack{a \to a/2 \\ b = 8a}]{} \quad C_1 = \frac{2\pi\epsilon_0 l_1}{\ln(8)}$$

$$\ln 8 = \ln(4 \cdot 2)$$
$$= \ln(4) + \ln(2)$$

$$\frac{C_0}{C_1} = 1 \quad \Rightarrow \quad \frac{2\pi\epsilon_0 l / \ln(4)}{2\pi\epsilon_0 l_1 / \ln(8)} = 1$$

$$\frac{l}{\ln(4)} = \frac{l_1}{\ln(8)}$$

$$\Rightarrow \quad l_1 = \frac{\ln(8)}{\ln(4)} l.$$

$$= \left[\frac{\ln(4)}{\ln(4)} + \frac{\ln(2)}{\ln(4)}\right] l$$

$$= \left[1 + \frac{1}{2}\right] l \quad \Rightarrow \quad l_1 = 1{,}5\, l$$